

Notes: d'Astous & Shore (JFE, 2024)

Human Capital Risk and Portfolio Choices: Evidence from University Admission Discontinuities

Contents

1	Big picture	2
2	Data and measurement	2
2.1	Danish admission discontinuities	2
2.2	Earnings process	3
2.3	Financial outcomes	3
2.4	Program-level versus individual-level measures	4
3	Models and empirical specifications	4
3.1	Effect of program on individual earnings process	4
3.2	Effect of earnings process on financial choices	4
3.3	Admission-discontinuity first stage	4
3.4	Regression-discontinuity controls	5
3.5	Portfolio-choice calibration	5
4	Identification and empirical design	5
4.1	Why the cutoff comparison is credible	5
4.2	Why ordinary least squares is biased	6
4.3	Threats addressed by the paper	6
5	Tables and figures	6
5.1	Figure 1 and Table 1: admission discontinuity and sample summary	6
5.2	Table 2 and Table 3: cohort averages and earnings first stage	7
5.3	Table 4 and Table 5: main portfolio effects and heterogeneity	8
5.4	Figure 2 and Table 6: robustness and calibration inputs	8
5.5	Figure 3: model fit	10
5.6	Selected appendix visuals: first stages, balance, and robustness	11
6	Short summary	13
7	Conclusion	13
7.1	Core conclusion	13
7.2	Economic intuition	14
7.3	Takeaway	14

1 Big picture

- **Question.** Does idiosyncratic, uninsurable labor-income risk cause individuals to reduce risk in their financial portfolios?
- **Theory behind the question.** Standard background-risk logic says that higher labor-income risk should make households less willing to hold risky financial assets, especially when labor risk is not tradable and is not strongly negatively correlated with asset returns.
- **Empirical problem.** Career choice and portfolio choice are jointly determined. Risk-tolerant people may choose riskier careers and also hold riskier portfolios, making raw correlations between earnings volatility and portfolio risk misleading.
- **Identification strategy.** The paper exploits discontinuities in the Danish centralized university admission system. Students just above a cutoff are admitted to their preferred “reach” program, while nearly identical students just below the cutoff usually attend a less-preferred “safety” program.
- **Key source of variation.** Different university programs generate different permanent earnings volatility. The admission discontinuity quasi-randomly assigns similar students to programs with different future earnings-risk profiles.
- **Main earnings first-stage result.** Program-level permanent earnings volatility passes through to individual permanent earnings volatility roughly one-for-one.
- **Main portfolio result.** Exogenously higher permanent earnings volatility reduces stock-market participation and reduces the risky-asset share conditional on participating.
- **Magnitude.** A one-standard-deviation increase in program-level permanent earnings volatility reduces stock-market participation by about 23% and reduces the risky-asset share by about 8%.
- **Raw correlations differ from causal estimates.** Non-instrumented estimates are much weaker or can have the wrong sign because students with greater risk tolerance appear to select into high-volatility fields and also hold riskier portfolios.
- **Model contribution.** A simple portfolio-choice model with risky labor income, modest participation costs, myopic behavior, and reasonable risk aversion can match the IV-estimated gradients.
- **Economic takeaway.** Human capital risk matters for household portfolio choice. The difficulty is not that theory is wrong, but that standard observational comparisons are biased by selection into careers and financial risk-taking.

2 Data and measurement

2.1 Danish admission discontinuities

- Students rank university programs and compete for seats using an application score.
- A program-specific cutoff determines admission. Applicants above the cutoff are admitted to the reach program; applicants below it usually enroll in the safety program.
- The paper focuses on applicants with a viable reach-safety pair, close to the reach cutoff and safely above the safety cutoff.
- The identifying sample contains 10,567 students applying in 1994–2006.

The basic assignment rule is

$$p_i = \begin{cases} r_i, & D_i \geq 0, \\ s_i, & D_i < 0, \end{cases} \quad (1)$$

where p_i is the program of enrollment, r_i is the reach program, s_i is the safety program, and D_i is the distance between the student's admission score and the reach-program cutoff.

Interpretation: The discontinuity creates quasi-random variation in program enrollment for students near the cutoff. This is the core design that separates causal effects of earnings risk from selection on risk preferences.

2.2 Earnings process

The paper estimates residual earnings using

$$w_{it} = c_t + \beta' X_{it}^w + u_{it}, \quad (2)$$

where w_{it} is log real annual earnings, c_t is a year effect, X_{it}^w contains controls, and u_{it} is excess earnings.

The earnings-growth residual is

$$\Delta u_{it} = u_{it} - u_{i,t-1}. \quad (3)$$

The paper studies several attributes of the earnings process:

- excess earnings: u ;
- excess earnings growth: Δu ;
- total earnings volatility: σ^2 ;
- transitory volatility: σ_τ^2 ;
- permanent volatility: σ_π^2 ;
- stock-market correlation of earnings: $\rho_{\Delta u, m}$.

The volatility decomposition uses moments of earnings-growth residuals, including

$$-\mathbb{E}[\Delta u_{it} \Delta u_{i,t-1}] \equiv \sigma_\tau^2, \quad \mathbb{E}[\Delta u_{it} \Delta u_{i,t+j}] \equiv \sigma_\pi^2. \quad (4)$$

Interpretation: Permanent volatility is the paper's main source of background risk. It is especially important because it affects lifetime labor-income uncertainty rather than only short-run noise.

2.3 Financial outcomes

The main financial outcomes are:

- stock-market participation: whether the person owns stocks;
- risky-asset share: stocks divided by financial wealth, conditional on stock-market participation;
- bond holdings, bank deposits, and other wealth variables for robustness.

Financial wealth is measured 8–12 years after enrollment, matching the period used to measure earnings outcomes.

Interpretation: The timing is important: admission affects program of study, program of study affects earnings risk after schooling, and portfolio choices are measured once individuals have entered adult financial life.

2.4 Program-level versus individual-level measures

- Individual-level earnings attributes are noisy because each individual has a limited post-school earnings history.
- Program-level averages reduce noise and capture the earnings-risk profile associated with a field of study.
- The core IV regressions use program-level earnings attributes as explanatory variables, instrumented by admission-discontinuity variation.

Interpretation: Program-level permanent volatility is the cleaner measure of the risk environment induced by university enrollment. The first-stage results show that this program-level measure predicts individual earnings risk.

3 Models and empirical specifications

3.1 Effect of program on individual earnings process

The first step asks whether program-level earnings processes transmit to individual earnings outcomes:

$$y_i = \delta y_{p_i} + \beta X_i + \epsilon_i, \quad (5)$$

where y_i is an individual earnings attribute and y_{p_i} is the analogous average attribute of the enrollment program.

Interpretation: When y is permanent volatility, δ measures whether enrolling in a high-volatility program causally increases the individual's own permanent earnings volatility.

3.2 Effect of earnings process on financial choices

The main portfolio equation is

$$SH_i = \Gamma(y_i \text{ or } y_{p_i}) + \beta X_i + \mu_i, \quad (6)$$

where SH_i is either stock-market participation or the risky-asset share.

Interpretation: The coefficient Γ measures the effect of earnings-process attributes on financial risk-taking. The coefficient on permanent earnings volatility is the central parameter.

3.3 Admission-discontinuity first stage

Because students select into programs, the paper instruments earnings attributes with predicted enrollment around the cutoff:

$$y_i \text{ or } y_{p_i} = \phi X_i + \chi Z_i + \xi_i. \quad (7)$$

The instruments include

$$\Pr(p_i = r_i \mid D_i) \times y_{r_i}, \quad \Pr(p_i = s_i \mid D_i) \times y_{s_i}. \quad (8)$$

Interpretation: The instrument is not simply being close to a cutoff. It is the discontinuous jump in the probability of attending the reach versus safety program, multiplied by the earnings-risk profile of those programs.

3.4 Regression-discontinuity controls

The estimation includes:

- a bandwidth of ± 0.5 around the reach cutoff in the baseline;
- distance-to-cutoff controls and interactions with quota fixed effects;
- reach and safety program-level controls;
- fixed effects for quota, cutoff, and calendar year;
- controls for parental wealth, income, education, student grades, age, and gender.

Interpretation: These controls make the comparison local to the admission cutoff and account for observable differences across programs and applicants.

3.5 Portfolio-choice calibration

The calibration uses a one-period portfolio problem with CRRA utility

$$u(C) = \frac{C^{1-\gamma} - 1}{1-\gamma}. \quad (9)$$

Final consumption is

$$C = W + Y, \quad (10)$$

where W is financial wealth and Y is labor income.

Financial wealth is

$$W = W_0 [\omega(1 + R) + (1 - \omega)(1 + R_f)], \quad (11)$$

where ω is the risky-asset share. Risky returns and labor income are lognormal:

$$r = \log(1 + R) \sim N(\bar{r}, \sigma_r^2), \quad y = \log(Y) \sim N(\bar{y}, \sigma_y^2). \quad (12)$$

Stock-market participation has a utility cost:

$$p_i = \begin{cases} \mathbb{E}u(C) - PUC, & \omega > 0, \\ \mathbb{E}u(C), & \omega = 0. \end{cases} \quad (13)$$

The certainty equivalent of participation costs is modeled as

$$CE(PUC) \sim N(\bar{p}, \sigma_p^2). \quad (14)$$

Interpretation: The model is used to see whether the causal gradients estimated by the IV design can be rationalized with plausible preference and participation-cost parameters.

4 Identification and empirical design

4.1 Why the cutoff comparison is credible

- Students very close to the reach cutoff are similar in grades and application characteristics.
- Crossing the cutoff sharply changes enrollment in the reach program and reduces enrollment in the safety program.
- Program cutoffs move across years, reducing the feasibility of strategic manipulation.
- Density and balance tests in the appendix support the validity of the cutoff design.

Interpretation: The admission cutoff acts like a local randomized assignment to programs with different earnings-risk profiles.

4.2 Why ordinary least squares is biased

- Risk-tolerant students may choose programs with higher earnings volatility.
- Risk-tolerant students may also prefer risky financial portfolios.
- Raw correlations therefore tend to obscure the negative causal effect of earnings volatility on risky-asset demand.

Interpretation: The paper's IV estimates are negative and economically large, while non-IV estimates are much weaker. This is exactly what selection on risk tolerance would predict.

4.3 Threats addressed by the paper

- **Average earnings:** The regressions control for excess earnings and earnings growth, not only volatility.
- **Human capital beta:** The design tests whether the relevant variation is idiosyncratic background risk rather than covariance with the stock market.
- **Financial wealth:** Robustness checks instrument for financial wealth to rule out wealth accumulation as the main channel.
- **Program characteristics:** Reach and safety fixed effects absorb fixed characteristics of program pairs.
- **Bandwidth choice:** Figure 2 shows robustness across alternative bandwidths around admission cutoffs.

Interpretation: The main claim is not merely that high-risk programs have low-risk portfolios. The claim is that quasi-random assignment to high-volatility programs causes later portfolio de-risking.

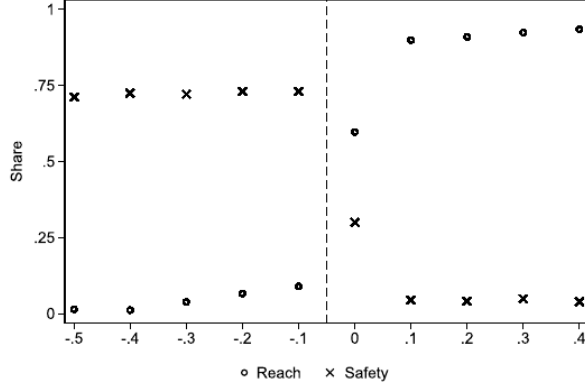
5 Tables and figures

5.1 Figure 1 and Table 1: admission discontinuity and sample summary

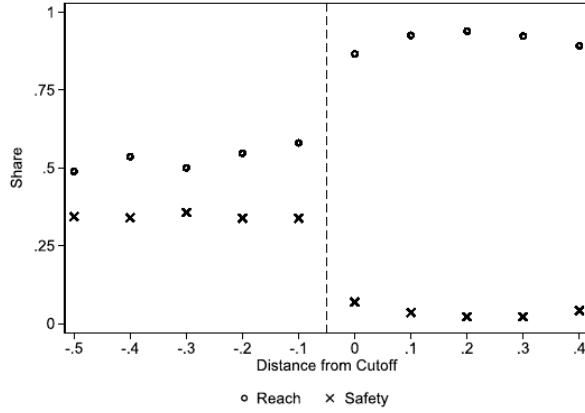
Read:

- Figure 1 shows discontinuous changes in enrollment probabilities at the reach-program cutoff.
- Just above the cutoff, admission to the reach program rises sharply; admission to the safety program falls sharply.
- Table 1 shows the descriptive statistics for all first-time applicants and the 10,567-student reach-safety analysis sample.
- The used sample is about 21 years old on average, has stronger scores than the full applicant population, and has similar stock-market participation of about 28%.
- Average permanent volatility of log annual earnings is about 0.015 in the used sample.

Economic message: The discontinuity creates the identifying variation, while Table 1 gives the key baseline moments used in both the empirical design and the calibration.



(a) Quota 1 Applications



(b) Quota 2 Applications

Fig. 1. Outcome of Applications. *Note:* This figure shows the share of students admitted to their reach or safety program. We pool all programs and years and present the distance between the application score and the admission cutoff for

(a) Figure 1: outcome of applications

Table 1
Descriptive statistics.

	All applications (321,090 obs.)		Used sample (10,567 obs.)	
	Mean	Std. dev	Mean	Std. dev
<i>I. Demographics</i>				
Age	21.5	3.9	21.1	3.0
Male share	37.2	-	38.8	-
Parental income (000's DKK)	297.4	157.3	329.3	173.3
Parental wealth (000's DKK)	353.6	751.7	442.1	860.1
Parental education (years)	12.7	3.2	13.4	3.2
<i>II. Education</i>				
Score	8.2	1.0	8.8	0.7
Nbr priorities ranked	2.2	1.4	3.1	1.4
Share got in reach	-	-	0.6	0.5
Rank of reach	-	-	1.2	0.6
Rank of safety	-	-	2.3	0.7
Rank of enrollment prog.	-	-	1.6	0.9
Distance from cutoff	-	-	-0.1	0.3
Share same field	-	-	0.3	0.5
Share same institution	-	-	0.6	0.5
<i>III. Earnings</i>				
Years until initial earnings	6.5	3.2	7.7	3.1
Initial earnings (000's DKK)	249.3	44.5	256.1	47.0
Average earnings (000's DKK)	280.4	141.2	282.6	149.4
PV human capital wealth (000's DKK)	5,593	2,836	5,661	3,006
Excess earnings: u	0.001	0.232	0.001	0.239
Excess earnings growth: Δu	0.007	0.078	0.016	0.085
Total vol.: σ^2	0.073	0.095	0.080	0.100
Trans. vol.: σ^2	0.029	0.059	0.032	0.063
Perm. vol.: σ^2	0.012	0.063	0.015	0.069
Stock corr.: $\rho_{\Delta u, u}$	0.062	0.256	0.054	0.244
<i>IV. Financial wealth</i>				
Share own stocks	0.28	-	0.28	-
Stocks (000's DKK) owns stocks	37.23	65.10	42.42	70.93
Share own bonds	0.08	-	0.08	-
Bonds (000's DKK) owns bonds	35.48	36.32	37.93	37.74
Share own bank deposits	1.00	-	1.00	-
Bank dep. (000's DKK) owns b. d.	65.69	88.70	72.80	93.84
Financial wealth (000's DKK)	78.79	110.9	87.85	119.5
PV h.c. wealth/total wealth	0.048	0.17	0.054	0.18
Stocks/Fin. wealth: RS	0.071	0.17	0.072	0.17
Stocks/Fin. wealth owns stocks	0.25	0.24	0.25	0.24

Note: This table presents descriptive statistics for the population of students applying to university for the first time in application years 1994 to 2006 (all applications; left), as well as the sample of 10,567 students who have a reach-safety programs pair in these application years (used sample; right). Panel I presents the demographics, measured at the time of application.

(b) Table 1: descriptive statistics

5.2 Table 2 and Table 3: cohort averages and earnings first stage

Read:

- Table 2 shows that reach and safety programs differ in earnings-process attributes, including permanent earnings volatility.
- The reach-safety permanent-volatility gap is about 0.003, and the cross-sectional standard deviation across programs is about 0.008.
- Table 3 shows that program-level permanent volatility passes through to individual permanent volatility with an IV coefficient of 1.00 and strong significance.
- Program-level total and transitory volatility also affect individual total and transitory volatility, although the permanent component is the main object of interest.
- Excess earnings pass through from program to individual outcomes, but excess earnings growth is not significantly affected in IV estimates.

Economic message: The first-stage result validates the design: attending a high-volatility program actually changes the individual's own future earnings risk.

Table 2
Cohort averages.

	All applications (321,090 obs.)		Used sample (10,567 obs.)		Reach	Safety	Gap
	Own	Enroll. cohort	Own	Enroll. cohort			
<i>A. Earnings process</i>							
Years until init. earn.	6.53 (3.20)	6.52 (2.97); (1.23)	7.67 (3.09)	7.16 (3.04); (1.20)	7.30 (3.01); (1.13)	6.97 (3.06); (1.29)	0.33 (1.11)
Init. earn. (000's DKK)	249.28 (44.54)	296.28 (41.12); (18.01)	253.97 (46.37)	227.63 (45.77); (17.49)	260.90 (47.58); (19.18)	253.74 (43.88); (14.87)	7.17 (17.93)
Earns earn.: μ	0.001 (0.232)	-0.002 (0.112); (0.092)	0.010 (0.238)	0.010 (0.223); (0.098)	0.010 (0.220); (0.103)	-0.003 (0.226); (0.091)	0.020 (0.093)
Earns earn. grow.: $\Delta \mu$	0.007 (0.078)	0.007 (0.083); (0.015)	0.022 (0.082)	0.013 (0.096); (0.017)	0.016 (0.093); (0.018)	0.009 (0.086); (0.015)	0.006 (0.017)
Total vol.: σ^2	0.073 (0.095)	0.072 (0.095); (0.018)	0.082 (0.099)	0.080 (0.103); (0.017)	0.083 (0.101); (0.018)	0.076 (0.100); (0.016)	0.007 (0.018)
Trans. vol.: σ_1^2	0.029 (0.059)	0.029 (0.058); (0.008)	0.033 (0.063)	0.031 (0.064); (0.008)	0.033 (0.062); (0.009)	0.030 (0.062); (0.008)	0.003 (0.009)
Perm. vol.: σ_2^2	0.012 (0.083)	0.012 (0.059); (0.008)	0.016 (0.069)	0.015 (0.066); (0.009)	0.016 (0.060); (0.009)	0.014 (0.062); (0.008)	0.003 (0.009)
Stock corr.: $\rho_{\Delta \mu, \sigma_2^2}$	0.062 (0.256)	0.041 (N.A.); (0.249)	0.008 (0.277)	0.043 (N.A.); (0.240)	0.039 (N.A.); (0.234)	0.048 (N.A.); (0.244)	-0.009 (0.296)
<i>B. Financial wealth</i>							
Share own stocks	0.282 (N.A.)	0.279 (0.437); (N.A.)	0.284 (N.A.)	0.280 (0.439); (N.A.)	0.280 (0.441); (N.A.)	0.279 (0.437); (N.A.)	0.0005 (0.082)
Stocks (000's DKK)	10.49 (38.40)	10.84 (36.70); (6.60)	12.06 (42.39)	12.03 (40.06); (6.69)	12.40 (41.39); (6.33)	11.57 (38.47); (7.05)	0.83 (6.40)
Bonds (000's DKK)	2.65 (13.63)	2.81 (12.84); (1.68)	3.04 (14.84)	3.08 (13.84); (1.38)	3.16 (14.20); (1.49)	2.96 (13.31); (1.72)	0.20 (1.47)
Bank dep. (000's DKK)	65.64 (88.69)	65.94 (87.97); (16.83)	72.74 (93.82)	71.80 (93.85); (16.44)	75.01 (96.36); (16.30)	68.19 (91.03); (16.22)	6.83 (15.55)
Stocks/Fin. wealth	0.071 (0.169)	0.072 (0.164); (0.033)	0.073 (0.171)	0.073 (0.168); (0.031)	0.074 (0.169); (0.027)	0.074 (0.167); (0.034)	-0.0009 (0.030)
Share own house	0.587 (N.A.)	0.572 (0.48); (N.A.)	0.511 (N.A.)	0.523 (0.487); (N.A.)	0.519 (0.488); (N.A.)	0.526 (0.488); (N.A.)	-0.007 (0.104)
House value (000's DKK)	450.07 (528.45)	446.07 (519.08); (103.00)	425.48 (555.69)	433.18 (545.94); (111.13)	439.23 (557.20); (110.24)	425.00 (537.20); (112.29)	14.22 (110.88)
Mortgage (000's DKK)	379.92 (440.03)	370.91 (428.23); (90.57)	350.18 (458.11)	352.76 (444.10); (98.64)	356.28 (452.90); (98.43)	347.59 (452.90); (98.90)	8.68 (97.74)

Note: This table presents descriptive statistics for the population of students applying to university for the first time in application years 1994 to 2006 (all applications).

5.3 Table 4 and Table 5: main portfolio effects and heterogeneity

Read:

- Table 4 is the main result. Program-level permanent earnings volatility reduces stock-market participation and the risky-asset share.
- The IV coefficient on program-level permanent volatility is about -6.36 for owning stocks and about -1.88 for stocks divided by financial wealth.
- The analogous non-IV estimates are much smaller and do not reproduce the same causal pattern.
- Table 5 shows heterogeneity. The effect of permanent earnings volatility is stronger when expected labor earnings are higher.
- The interaction results are consistent with the idea that labor-income risk matters more when human capital is a larger part of total wealth.

Economic message: Higher uninsurable earnings risk causes individuals to take less financial risk. The heterogeneity reinforces the background-risk interpretation.

Table 4
Effects of earnings processes on individual financial holding.

	Program-level explanatory variables		Individual-level explanatory variables	
	Own stocks (1)	Stocks/Fin. wealth (2)	Own stocks (3)	Stocks/Fin. wealth (4)
<i>A. IV estimation</i>				
Earns earnings: μ	0.51*** (0.20)	0.13** (0.06)	0.33 (0.30)	0.15 (0.16)
Trans. vol.: σ_1^2	0.08 (1.79)	-0.027 (0.56)	0.089 (2.45)	0.043 (1.56)
Perm. vol.: σ_2^2	-6.36*** (2.25)	-3.56*** (0.69)	-2.05 (0.87)	-2.05 (1.31)
<i>B. Non-IV estimation</i>				
Earns earnings: μ	0.37*** (0.35)	0.11*** (0.016)	0.20*** (0.01)	0.049*** (0.003)
Trans. vol.: σ_1^2	-0.06 (0.35)	-0.016 (0.57)	-0.019 (0.57)	-0.023 (0.57)
Perm. vol.: σ_2^2	-0.57 (0.69)	-0.15 (0.14)	-0.073 (0.09)	-0.021 (0.02)

Note: This table presents the results of estimating equation (3). Each column presents the result from either the second stage of an IV regression (Panel A) or from a non-instrumented regression (Panel B). We use program- and individual-level explanatory variables in the left and right panels, respectively. The outcome variable for each regression is the student's own estimate of the variable studied, while the explanatory variables are instrumented using the reach and safety program-level instruments in equation (3). Stock market participation is estimated with a probit model, from which we report the marginal effects of participation, while the share of risky assets in the portfolio is estimated with a tobit model, from which we report marginal effects for the share of risky assets, conditional on participation. In both panels, we use the sample of 10,567 students who have a reach-safety program pair in application years 1994 to 2006 and present the results for a bandwidth of ± 0.5 around the admission cutoff for the reach program. Earnings are assumed to follow the process described in equation (2). The definitions of the different attributes of the earnings process follow from this equation and are averaged over the period 8 to 12 years after enrollment. Financial wealth measures are also averaged over the period 8 to 12 years after enrollment. The IV estimation controls for the reach and safety averages of the program-level variables being instrumented, admission system (quota 1 or 2) fixed effects, distance from the admission cutoff fixed effects and their interaction with admission quota fixed effects, the probability of being accepted given last year's admission cutoff and its interaction with quota fixed effects, and calendar year FE. The OLS results control for parents' wealth, income, and education, students' grade, age, and gender, and calendar year FE. Standard errors clustered at the reach-program-year level are presented in parentheses. ***, **, and * represent significance at the 1, 5 and 10 percent level, respectively.

Table 3
Effects of program-level earnings processes on individuals' earnings process.

	Individual-level outcomes				
	Earns earnings: μ	Earns earn. growth: $\Delta \mu$	Total vol.: σ^2	Trans. vol.: σ_1^2	Perm. vol.: σ_2^2
<i>A. IV estimation</i>					
Prog.-lvl expl. variable	0.53*** (0.23)	0.26 (0.23)	0.52* (0.20)	0.55* (0.30)	1.00*** (0.36)
<i>B. OLS estimation</i>					
Prog.-lvl expl. variable	0.78*** (0.02)	0.54*** (0.06)	0.86*** (0.07)	1.02*** (0.10)	0.88*** (0.11)

Note: This table presents the results of estimating equation (3) by IVs (Panel A) and by OLS (Panel B). Each cell presents the result from either a second stage of an IV regression or from an OLS regression. The outcome variable for each regression is the student's own estimate of the variable studied, while the explanatory variables are program-level average outcomes. In Panel A, each program-level explanatory variable is instrumented using the reach and safety program-level instruments

Table 5
Effects of program-level earnings processes on individuals' financial holding: heterogeneous effects.

Characteristic for interaction	Own stocks					Stocks/Financial wealth				
	Own stocks (1)	Earns earnings: μ (2)	Fin. wealth (3)	Fin. wealth/Fin. wealth (4)	Age (5)	Own stocks (6)	Earns earnings: μ (7)	Fin. wealth (8)	Fin. wealth/Fin. wealth (9)	Age (10)
<i>A. IV estimation</i>										
Earns earnings: μ	0.51*** (0.20)	0.47*** (0.26)	0.36 (0.34)	0.54*** (0.18)	0.28 (0.28)	0.40 (0.11)	0.11*** (0.04)	0.14*** (0.05)	0.13* (0.07)	0.16*** (0.03)
Trans. vol.: σ_1^2	0.08 (1.79)	0.05 (1.84)	0.05 (1.81)	0.08 (2.01)	0.08 (1.84)	-0.49 (15.18)	-0.04 (0.84)	0.11 (0.80)	0.03 (0.76)	-0.05 (4.51)
Perm. vol.: σ_2^2	-6.36*** (2.25)	-6.36*** (3.17)	-3.51 (2.83)	-6.81*** (2.96)	-10.30 (17.60)	-1.80*** (0.91)	-1.80*** (0.70)	-2.38*** (0.75)	-2.38*** (0.73)	-4.18 (3.20)
Earns earnings: $\mu \times$ characteristic										
Trans. vol.: $\sigma_1^2 \times$ characteristic	2.73 (0.57)	0.29 (0.57)	-14.30 (16.49)	-0.09 (0.58)	-0.04 (0.58)	0.00 (0.40)	0.13 (0.30)	-0.05 (0.30)	0.03 (0.30)	-0.00 (0.00)
Perm. vol.: $\sigma_2^2 \times$ characteristic	-2.17** (0.98)	-2.11 (0.98)	-26.67 (13.10)	-0.02 (0.87)	-0.02 (0.87)	-8.13*** (2.86)	-8.16*** (0.86)	-9.96 (4.63)	-9.96 (4.63)	-0.38 (0.30)
<i>B. Non-IV estimation</i>										
Earns earnings: μ	0.37*** (0.06)	0.27*** (0.06)	0.20*** (0.07)	0.20*** (0.06)	0.23 (0.47)	0.11*** (0.02)	0.082*** (0.02)	0.12*** (0.02)	0.10*** (0.02)	0.02 (0.16)
Trans. vol.: σ_1^2	-0.25 (0.57)	-0.25 (0.62)	-0.12 (0.62)	-0.48 (0.58)	-0.62 (0.40)	-0.04 (0.15)	0.15 (0.17)	-0.02 (0.15)	-0.02 (0.15)	-2.79* (1.20)
Perm. vol.: σ_2^2	-0.69 (0.63)	-0.57 (0.63)	-0.35 (0.62)	-0.58 (0.67)	-0.35 (0.40)	0.14 (0.18)	0.11 (0.19)	0.12 (0.12)	0.12 (0.12)	0.79 (0.64)
Earns earnings: $\mu \times$ characteristic										
Trans. vol.: $\sigma_1^2 \times$ characteristic	5.39** (2.16)	0.04 (0.65)	2.60 (0.65)	0.04 (0.65)	0.31 (0.51)	1.39** (0.56)	0.03 (0.36)	0.03 (0.36)	0.13* (0.06)	0.13* (0.06)
Perm. vol.: $\sigma_2^2 \times$ characteristic	-3.92 (1.80)	-0.53 (0.67)	-2.01 (0.73)	-0.01 (0.73)	-0.01 (0.73)	-6.74 (0.67)	-0.12 (0.12)	-0.38 (0.12)	-0.38 (0.12)	-0.09 (0.09)
Characteristic sample average	0.601	0.58	0.65	0.55	21.1	0.003	0.88	0.05	0.05	21.1

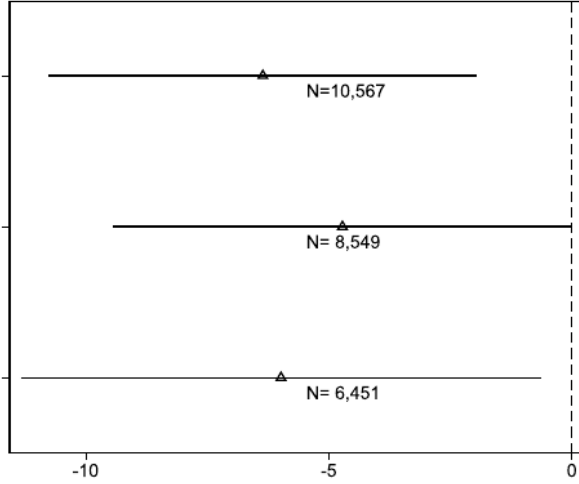
Note: This table presents the heterogeneous effects of our outcome to measures of expected human capital, financial wealth, and age. Specifically, it presents the results of estimating equation (3) by IV (Panel A) and directly, without any instrument (Panel B). Each column presents the result from either a second stage of an IV regression or from a non-instrumented regression. The outcome variable for each regression is the student's own estimate of the variable studied, while the explanatory variables are program-level average outcomes. Stock market participation is estimated with a probit model, from which we report the marginal effects of participation, while the share of risky assets in the portfolio is estimated with a tobit model, from which we report the marginal effects of participation, while the share of risky assets, conditional on participation. In Panel A, each program-level explanatory variable is instrumented using the reach and safety program-level instruments in equation (3). In both panels, we use the sample of 10,567 students who have a reach-safety program pair in application years 1994 to 2006 and present the results for a bandwidth of ± 0.5 around the admission cutoff for the reach program. Earnings are assumed to follow the process described in equation (2). The definitions of the different attributes of the earnings process follow from this equation and are averaged over the period 8 to 12 years after enrollment. Average earnings, financial wealth and total wealth are averaged over the period 8 to 12 years after enrollment. Age is measured at the time of enrollment. A part from the variables presented, the IV estimation always controls for the reach and safety averages of the program-level variables being instrumented, admission system (quota 1 or 2) fixed effects, distance from the admission cutoff fixed effects and their interaction with admission quota fixed effects, and calendar year FE. The OLS results control for parents' wealth, income, and education, students' grade, age, and gender, and calendar year FE. Standard errors clustered at the reach-program-year level are presented in parentheses. ***, **, and * represent significance at the 1, 5 and 10 percent level, respectively.

5.4 Figure 2 and Table 6: robustness and calibration inputs

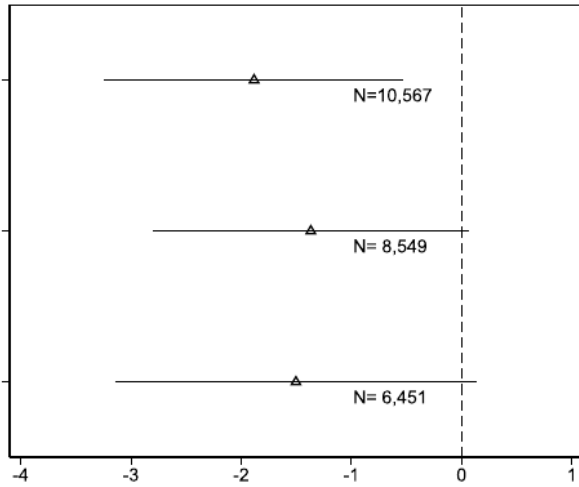
Read:

- Figure 2 shows that the main IV results are robust across alternative bandwidths around the admission cutoff.
- The permanent-volatility effect remains negative for both stock-market participation and risky-asset share.
- Table 6 reports the inputs and calibration targets.
- The calibration uses average financial wealth of 87,848 DKK, average annual earnings of 282,578 DKK, risky share of 0.25, and participation share of 0.28.
- Calibrated parameters include risk aversion of 12.35, a human capital multiplier of 1.10, and modest participation-cost moments.

Economic message: The reduced-form effect is not driven by a single bandwidth choice. The calibrated model shows that the causal slopes can be explained with reasonable parameters once participation costs and myopia are included.



(a) Stock Market Participation



(b) Risky asset share

alternative Bandwidths. *Note:* This figure presents the robustness of IV results (columns 2 and 4 of Table 4) to the selection of different bandwidths. We use the sample of 10 567 students who have a reach-safety

(a) Figure 2: alternative bandwidths

Table 6
Inputs, moments and parameters used in the calibration.

Parameter	Value	Source
<i>A. Individual inputs</i>		
Financial wealth (DKK)	87,848	Table 1
Annual earnings (DKK)	282,578	Table 1
Average permanent vol. of log annual earnings	0.015	Table 1
<i>B. Financial market inputs</i>		
Annual risk-free rate (%)	3.30	Nielsen and Risager (2001)
Annual equity risk premium (%)	2.87	Dimson et al. (2008)
Annual st. dev. of risky asset return (%)	19.85	Dimson et al. (2008)
<i>C. Target moments</i>		
Stock market participation share	0.28	Table 1
Gradient of the participation w.r.t earnings vol	0.22 to 0.34	Column (2) of Table 5
Risky asset share	0.25	Table 1
Gradient of the risky share w.r.t earnings vol	0.23 to 0.27	Column (7) of Table 5
<i>D. Calibrated parameters: Baseline model</i>		
Risk aversion (γ)	12.35	
Human capital multiplier	1.10	
Mean of CE of participation cost $\bar{p}$ (DKK)	389.70	
Std. dev. of CE of participation cost σ_p (DKK)	161.31	

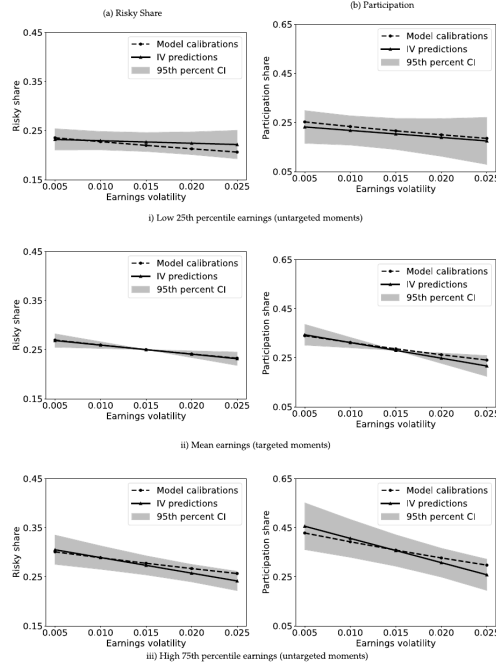
Note: This table presents the value of the inputs, moments and parameters used in our calibration. Values presented in panels A and C are obtained from averages of the reach-safety sample of our data. These financial outcomes are averaged over the period 8 to 12 years after enrollment. Financial wealth is defined as the sum of bank deposits, stocks, and bonds. In the calibration, we vary the average permanent volatility of log annual earnings.

(b) Table 6: calibration inputs and parameters

5.5 Figure 3: model fit

Read:

- Figure 3 compares model predictions with IV-implied moments for risky share and stock-market participation.
- At average earnings, the model matches the targeted slope of risky asset share with respect to earnings volatility.
- The empirical results imply a decline in risky share from about 26.9% at low earnings volatility to about 23.2% at high earnings volatility.
- The model also matches the qualitative pattern that the earnings-risk effect is stronger at higher earnings



levels.

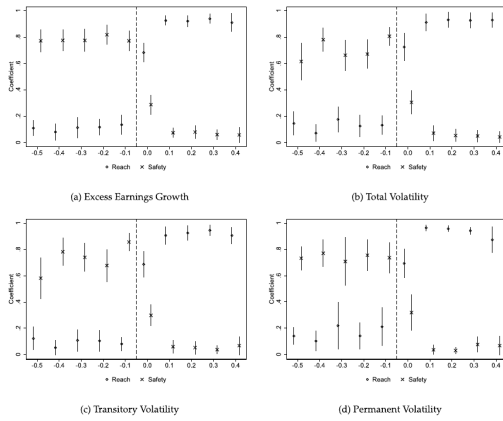
Economic message: The model helps translate the causal estimates into portfolio-choice primitives: background risk, risk aversion, participation costs, and effective human-capital valuation.

5.6 Selected appendix visuals: first stages, balance, and robustness

Read:

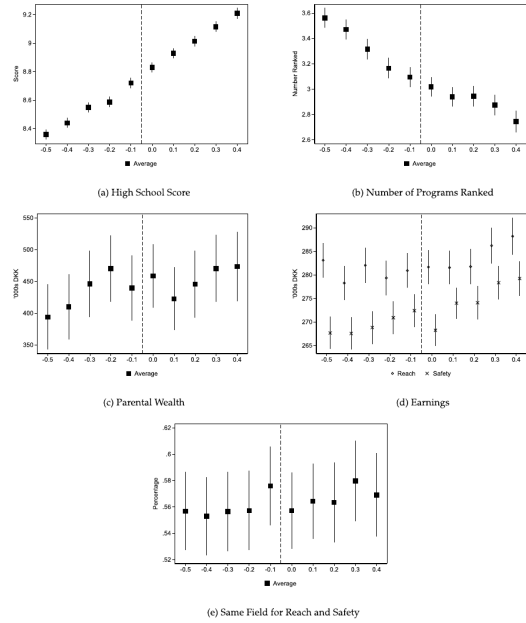
- Figure A.2 graphically illustrates the first-stage discontinuities in earnings attributes.
- Figure A.3 shows covariate balance around the admission cutoff.
- Tables A.3 and A.4 report strong first-stage estimates for the main earnings-process and financial-holding specifications.
- Tables A.8 and A.9 show robustness of the main Table 4 results to alternative specifications.

Economic message: The appendix evidence supports the credibility of the RD-IV design and shows that the portfolio-risk result is not an artifact of a narrow baseline specification.



A.2. First-stage Results. Note: This figure provides a graphical illustration of the first-stage earnings and earnings gap results. We use the sample of students who applied through the Quota 1 system of admission and have a reach-safety program pair in application years 1994 to 2006. We pool all programs and present the distance between the application score and the admission cutoff for the reach program on the x-axis. The results show the coefficients estimated separately for the reach and safety programs using 10 dummy bins centered around the admission cutoff. Each student has exactly one dummy bin variable 1, with the rest equal to zero. In each panel, the dependent variable is a attribute of the earnings process of enrollees in the program of admission, and v_i coefficients of earnings in the reach and safety programs interacted with distance from cutoff dummies. Earnings are assumed to follow the process described in equation (2). The definitions of the different attributes of the earnings process follow from this equation and are averaged over the period 8 to 12 year rollout. Point estimates are given along with the 95% confidence interval.

(a) Figure A.2: first-stage results



A.3. Covariate Balance. Note: This figure presents averages of the covariates used in the analysis by application score bins. We use the sample of 10,567 students who have a reach-safety program pair in application years 1994 to 2006. We pool all programs and years and present the distance between the application score and the admission cutoff of the reach program on the x-axis. Point estimates are given along with the 95% confidence interval.

References

- Barth, D., Shore, S.H., Jensen, S.T., 2017. Identifying idiosyncratic career taste as with income risk. *Quant. Econ.* 8, 553–587.
 Bertini, C.C., Hallsson, M., 1997. Precautionary portfolio behavior from a life-cycle perspective. *J. Econ. Dyn. Control* 21, 1511–1542.
 Berman, S., Jamson, T., Parfou, C., Walden, J., 2012. Hedging labor income. *Finance Econ.* 101, 623–639.
 Blundell, R., Pissaferr, L., Preston, I., 2008. Consumption inequality and participation. *Am. Econ. Rev.* 98, 1887–1921.
 Boyle, J., Martin, B.P., 2005. 1995. *1995 Annual Report*. University of Wisconsin-Madison, 1995.

(b) Figure A.3: covariate balance

Table A.3
Validity of instruments: first stage estimations of Table 3: effects of program-level earnings processes on individual earnings processes.

	Individual-level outcomes				
	Excess earnings: α	Excess earnings growth: Δv	Total vol.: σ^2	Trans. vol.: σ^2_t	Perm. vol.: σ^2_p
$P(\text{reach}) = \alpha_1 D_i \times u_i$	0.93*** (0.03)				
$P(\text{safety}) = \alpha_2 D_i \times u_i$	1.04*** (0.04)				
$P(\text{reach}) = \alpha_1 D_i \times \Delta v_i$		0.89*** (0.04)			
$P(\text{safety}) = \alpha_2 D_i \times \Delta v_i$		1.00*** (0.05)			
$P(\text{reach}) = \alpha_1 D_i \times \sigma^2_i$			0.88*** (0.05)		
$P(\text{safety}) = \alpha_2 D_i \times \sigma^2_i$			0.94*** (0.06)		
$P(\text{reach}) = \alpha_1 D_i \times \sigma^2_{it}$				0.96*** (0.04)	
$P(\text{safety}) = \alpha_2 D_i \times \sigma^2_{it}$				0.98*** (0.07)	
$P(\text{reach}) = \alpha_1 D_i \times \sigma^2_{ip}$					0.81*** (0.07)
$P(\text{safety}) = \alpha_2 D_i \times \sigma^2_{ip}$					0.96*** (0.07)
F-test of excluded instruments	595	298	221	302	95
Hansen over identification p-value	0.11	0.43	0.80	0.26	0.08

Note: This table presents the first stage estimation results from equation (7). Each column presents the result from the first stage of an IV regression. We use the sample of 10,567 students who have a reach-safety program pair in application years 1994 to 2006 and present the results for a bandwidth of ± 0.5 around the admission cutoff for the reach program. Earnings are assumed to follow the process described in equation (2). The definitions of the different attributes of the earnings process follow from this equation and are averaged over the period 8 to 12 years after enrollment. The outcome variable for each regression is the student's own estimate of the variable studied, while the explanatory variable is the analogous program-level average outcome. In all cases, the IV estimation controls for distance from cutoff fixed effects (bin FE), the reach and safety average outcome from the program-level variables being instrumented, calendar year FE, and the probability of being accepted given last year's admission cutoff. Standard errors clustered at the reach-program-year level are presented in parentheses. ***, **, and * represent significance at the 1, 5 and 10 percent level, respectively.

(a) Table A.3: first-stage earnings estimates

Table A.4
Validity of instruments: first stage estimations of Table 4: effects of earnings processes on individual financial holding.

	Enrollment program-level outcomes			Individual-level outcomes		
	Excess earnings: α	Trans. vol.: σ^2_t	Perm. vol.: σ^2_p	Excess earnings: α	Trans. vol.: σ^2_t	Perm. vol.: σ^2_p
$P(\text{reach}) = \alpha_1 D_i \times u_i$	0.94*** (0.03)	0.0037 (0.003)	0.0062** (0.003)	0.47*** (0.10)	0.01 (0.03)	-0.056** (0.03)
$P(\text{safety}) = \alpha_2 D_i \times u_i$	1.03*** (0.04)	-0.0075 (0.005)	0.003 (0.004)	0.47*** (0.13)	-0.0037 (0.04)	0.0078 (0.04)
$P(\text{reach}) = \alpha_1 D_i \times \sigma^2_{it}$	-0.16 (0.30)	0.98*** (0.04)	0.0021 (0.03)	0.4 (1.00)	0.76** (0.32)	0.12 (0.33)
$P(\text{safety}) = \alpha_2 D_i \times \sigma^2_{it}$	-0.30 (0.42)	0.95*** (0.08)	-0.018 (0.08)	0.54 (1.21)	0.16 (0.44)	0.33 (0.37)
$P(\text{reach}) = \alpha_1 D_i \times \sigma^2_{ip}$	-0.35 (0.32)	-0.0007 (0.05)	0.76*** (0.08)	0.88 (1.23)	1.36** (0.48)	0.74 (0.42)
$P(\text{safety}) = \alpha_2 D_i \times \sigma^2_{ip}$	-0.13 (0.27)	0.94*** (0.05)	0.13 (0.08)	0.11 (1.31)	0.23 (0.41)	0.11 (0.50)
F-test of excluded instruments	293	199	53	9	2	2
Hansen over identification p-value	0.79			0.80		

Note: This table presents the first stage estimation results from equation (7). Each column presents the result from the first stage of an IV regression. We use the sample of 10,567 students who have a reach-safety program pair in application years 1994 to 2006 and present the results for a bandwidth of ± 0.5 around the admission cutoff for the reach program. Earnings are assumed to follow the process described in equation (2). The definitions of the different attributes of the earnings process follow from this equation and are averaged over the period 8 to 12 years after enrollment. In the left panel, the outcome variable for each regression is the student's enrollment program-level outcome. In the right panel, the outcome variable for each regression is the student's own estimate of the variable studied. In both cases, the instruments are reach and safety program-level average variables. The IV estimation controls for the reach and safety averages of the program-level variables being instrumented, admission system (quota 1 or 2) fixed effects, distance from the admission cutoff fixed effects and their interaction with admission quota fixed effects, the probability of being accepted given last year's admission cutoff and its interaction with quota fixed effects, and calendar year FE. The OLS results control for parents' wealth, income, and education, students' grade, age, and gender, and calendar year FE. Standard errors clustered at the reach-program-year level are presented in parentheses. ***, **, and * represent significance at the 1, 5 and 10 percent level, respectively.

Table A.5
Validity of instruments: weak IV robust tests: effects of earnings processes on individual financial holding.

(b) Table A.4: first-stage financial estimates

Table A.8
Effects of program-level earnings processes on individuals' financial holding robustness to specifications.

		Individual-level outcomes											
		Own stocks						Stocks/Financial wealth					
		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
A. IV estimation													
Excess earnings: μ		0.54*** (0.19)	0.51*** (0.20)	0.52*** (0.21)	0.48*** (0.19)	0.42*** (0.16)	0.16*** (0.06)	0.15*** (0.06)	0.14*** (0.06)	0.14*** (0.06)	0.11*** (0.05)		
Trans. vol.: σ_1^2					-0.39 (1.79)	-0.048 (1.80)	-0.037 (1.68)	-0.041 (1.54)	-0.26 (0.56)	-0.036 (0.56)	-0.06 (0.53)	-0.26 (0.47)	-0.13 (0.48)
Perm. vol.: σ_2^2		-0.47 (2.14)	-0.21*** (2.25)	-0.36*** (2.25)	-0.50*** (2.42)	-1.18*** (2.19)	-1.04 (1.83)	-1.06 (0.66)	-1.89*** (0.69)	-2.03*** (0.76)	-1.73*** (0.66)	-1.31*** (0.51)	
Excess earn. growth: $\Delta\mu$													
Previous year admin. prob.	YES	YES	YES	YES	YES	NO	YES	YES	YES	YES	YES	YES	NO
Previous year admin. prob. x reach and safety controls	YES	YES	YES	YES	YES	NO	YES	YES	YES	YES	YES	YES	NO
OLS Controls	NO	NO	NO	NO	YES	NO	NO	NO	NO	NO	YES	NO	NO
B. Non-IV estimation													
Excess earnings: μ		0.37*** (0.06)	0.37*** (0.06)	0.35*** (0.06)	0.37*** (0.06)	0.37*** (0.06)	0.11*** (0.02)	0.11*** (0.02)	0.10*** (0.02)	0.11*** (0.02)	0.11*** (0.02)	0.11*** (0.02)	
Trans. vol.: σ_1^2			-0.35 (0.37)	-0.41 (0.38)	-0.35 (0.37)	-0.35 (0.37)	-0.037 (0.13)	-0.042 (0.13)	-0.037 (0.13)	-0.042 (0.13)	-0.037 (0.13)		
Perm. vol.: σ_2^2		3.01*** (0.53)	0.59 (0.64)	0.69 (0.65)	0.41 (0.76)	0.69 (0.65)	0.69 (0.65)	0.82*** (0.15)	0.13 (0.18)	0.14 (0.18)	0.12 (0.21)	0.14 (0.18)	0.14 (0.18)
Excess earn. growth: $\Delta\mu$													
				</									

Note: This table presents the robustness to different specifications for our main results presented in the left panel of Table 4. Specifically, it presents the results of estimating equation (5) by IVs (Panel A) and directly, without any instruments (Panel B). Each column presents the result from either a second stage of an IV regression or from a non-instrumented regression. The outcome variable for each regression is the student's own estimate of the variable studied, while the explanatory variables are program-level average outcomes. Stock market participation is estimated with a probit model, from which we report the marginal effects of participation, while the share of risky assets in the portfolio is estimated with a tobit model, from which we report marginal effects for the share of risky assets, conditional on participation. In Panel A, each program-level explanatory variable is instrumented using the reach and safety program-level instruments in equation (8). In both panels, we use the sample of 10,567 students who have a reach-safety programs pair in application years 1994 to 2006 and present the results for a bandwidth of ± 0.5 around the admission cutoff for the reach program. Earnings are assumed to follow the process described in equation (2). The definitions of the different attributes of the earnings process follow from this equation and are averaged over the period 8 to 12 years after enrollment. Financial wealth is averaged over the period 8 to 12 years after enrollment. A part from the variations presented, the IV estimation always controls for the reach and safety averages of the program-level variables being instrumented, admission system (quota 1 or 2) fixed effects, distance from the admission cutoff fixed effects and their interaction with admission quota fixed effects, and calendar year FE. The OLS results control for parents' wealth, income, and education, students' grade, age, and gender, and calendar year FE. Standard errors clustered at the reach-program-year level are presented in parentheses. ***, **, and * represent significance at the 1, 5 and 10 percent level, respectively.

(a) Table A.8: robustness for stock ownership

Table A.9
Effects of individuals' earnings processes on individuals' financial holding robustness to specifications.

	Individual-level outcomes											
	Own stocks						Stocks/Financial wealth					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
A. IV estimation												
Excess earnings: μ		0.37 (0.27)	0.33 (0.30)	0.19 (0.26)	0.37 (0.31)	0.33 (0.29)		0.18 (0.16)	0.15 (0.16)	0.12 (0.28)	0.18 (0.18)	0.13 (0.33)
Trans. vol.: σ_1^2			0.089 (2.45)	0.83 (1.99)	-0.088 (2.29)	-0.092 (4.14)			0.043 (1.56)	0.96 (2.04)	-0.065 (1.49)	0.57 (3.78)
Perm. vol.: σ_2^2	-2.78** (1.18)	-3.59*** (0.72)	-3.56*** (0.87)	-3.27*** (1.03)	-3.38*** (0.94)	-3.93*** (1.36)	-1.34 (0.85)	-2.12** (1.03)	-2.05 (1.31)	-3.23 (2.22)	-2.07 (2.23)	-2.64 (4.58)
Excess earn. growth: $\Delta\mu$							1.97* (1.09)					
Previous year admin. prob.	YES	YES	YES	YES	YES	NO	YES	YES	YES	YES	YES	NO
Previous year admin. prob. x reach and safety controls	YES	YES	YES	YES	YES	NO	YES	YES	YES	YES	YES	NO
OLS Controls	NO	NO	NO	NO	YES	NO	YES	NO	NO	YES	NO	YES
B. Non-IV estimation												
Excess earnings: μ		0.20*** (0.02)	0.20*** (0.02)	0.20*** (0.02)	0.20*** (0.02)	0.20*** (0.02)		0.049*** (0.01)	0.049*** (0.01)	0.049*** (0.01)	0.049*** (0.01)	0.049*** (0.01)
Trans. vol.: σ_1^2			-0.011 (0.07)	-0.018 (0.08)	-0.01 (0.07)	-0.011 (0.07)			-0.0047 (0.02)	-0.0042 (0.02)	-0.0047 (0.02)	-0.0047 (0.02)
Perm. vol.: σ_2^2	0.15** (0.06)	0.087 (0.06)	0.088 (0.06)	0.085 (0.06)	0.088 (0.06)	0.088 (0.06)	0.037** (0.02)	0.02 (0.02)	0.02 (0.02)	0.02 (0.02)	0.02 (0.02)	0.02 (0.02)
Excess earn. growth: $\Delta\mu$							0.026 (0.06)			-0.0019 (0.02)		

Note: This table presents the robustness to different specifications for our main results presented in the right panel of Table 4. Specifically, it presents the results of estimating equation (5) by IVs (Panel A) and directly, without any instruments (Panel B). Each column presents the result from either a second stage of an IV regression or from a non-instrumented regression. The outcome variable for each regression is the student's own estimate of the variable studied, while the explanatory variables are individual-level average outcomes. Stock market participation is estimated with a probit model, from which we report the marginal effects of participation, while the share of risky assets in the portfolio is estimated with a tobit model, from which we report marginal effects for the share of risky assets, conditional on participation. In Panel A, each individual-level explanatory variable is instrumented using the reach and safety program-level instruments in equation (8). In both panels, we use the sample of 10,567 students who have a reach-safety programs pair in application years 1994 to 2006 and present the results for a bandwidth of ± 0.5 around the admission cutoff for the reach program. Earnings are assumed to follow the process described in equation (2). The definitions of the different attributes of the earnings process follow from this equation and are averaged over the period 8 to 12 years after enrollment. Financial wealth is averaged over the period 8 to 12 years after enrollment. A part from the variations presented, the IV estimation always controls for the reach and safety averages of the program-level variables being instrumented, admission system (quota 1 or 2) fixed effects, distance from the admission cutoff fixed effects and their interaction with admission quota fixed effects, and calendar year FE. The OLS results control for parents' wealth, income, and education, students' grade, age, and gender, and calendar year FE. Standard errors clustered at the reach-program-year level are presented in parentheses. ***, **, and * represent significance at the 1, 5 and 10 percent level, respectively.

(b) Table A.9: robustness for risky-asset share

6 Short summary

- The paper studies whether labor-income risk causally affects household portfolio risk.
- The empirical setting is the Danish university admission system, where program cutoffs create quasi-random assignment to reach versus safety programs.
- The analysis sample is 10,567 students with reach-safety program pairs and data on earnings and financial wealth 8–12 years later.
- Program-level permanent earnings volatility causes individual permanent earnings volatility roughly one-for-one.
- Higher permanent earnings volatility causes lower stock-market participation and lower risky-asset shares.
- A one-standard-deviation increase in permanent earnings volatility reduces participation by about 23% and risky-asset share by about 8%.
- Raw correlations do not reveal this relationship because risk-tolerant students select into riskier programs and riskier portfolios.
- The estimates are robust to alternative bandwidths, program controls, wealth controls, human-capital beta controls, and alternative earnings-process assumptions.
- A simple calibrated portfolio-choice model can match the causal gradients using modest participation costs, myopic behavior, and reasonable risk aversion.

7 Conclusion

7.1 Core conclusion

d'Astous and Shore show that exogenous increases in labor-income risk cause people to take less risk in financial markets. The causal evidence comes from Danish university admission discontinuities that assign similar students to programs with different future earnings-risk profiles.

7.2 Economic intuition

When labor income is risky and not easily insured, it becomes a background risk. A person with more human-capital risk has less capacity to bear stock-market risk. Therefore, even if the stock market risk premium is unchanged, the optimal financial portfolio becomes safer. The key empirical challenge is separating this effect from selection on risk preferences; the university cutoff design solves this problem locally.

7.3 Takeaway

The paper provides clean causal evidence for a central prediction of portfolio theory: human capital risk affects financial portfolios. It also shows why simple correlations can be misleading. People who like risk may choose both volatile careers and risky portfolios, masking the negative causal effect of earnings risk on stock-market participation and risky-asset shares.